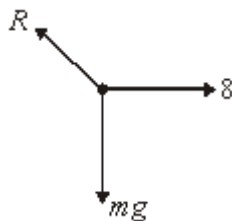


Mark schemes

Q1.

(a)



Correct force diagram with labels

B1

1

(b) $R \cos 30^\circ = 8$

Resolving horizontally to get two terms

M1

Correct equation

A1

$$R = \frac{8}{\cos 30^\circ} = 9.24$$

ag *Correct answer from correct working
(other methods: M1 A1 if correct)*

A1

3

(c) $R \cos 60 = 9.8m$

*Resolving horizontally to get two terms,
with 8 not included*

M1

Correct equation

A1

$$m = \frac{8 \cos 60^\circ}{9.8 \cos 30^\circ} = 0.47$$

Correct m to 2 sf

A1

3

*(Resolving perpendicular to the plane: M1A1
for equation and A1 for final answer)*

[7]

Q2.

(a) $R = 20 \times 9.8 = 196$

cao

B1

1

(b) $F \leq 0.3 \times 196 = 58.8$

Using 0.3×196

M1

If $P = 80$, $F = 58.8$

58.8 as answer

A1

If $P = 40$, $F = 40$

40 as answer

A1

3

(c) $P - 58.8 = 20 \times 0.8$

Three term equation of motion including 58.8

M1

Correct equation

A1

$P = 74.8$

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A1

Correct P

A1

3

(d) $a = \frac{-58.8}{20} = -2.94$

Use of $F = ma$ with ± 58.8

M1

Correct acceleration with a negative sign

A1

$0^2 = 6^2 + 2 \times (-2.94)s$

Use of $v^2 = u^2 + 2as$ with $v = 0$

m1 A1

$$s = \frac{36}{5.88} = 6.12$$

Correct distance

A1

5

[12]

Q3.

(a) $0.5 = \frac{1}{2} \times a \times 4$

Use of $s = ut + \frac{1}{2} at^2$ with $u = 0$

M1

$$a = 0.25$$

Correct acceleration

A1

2

(b) $6 \times 9.8 - T = 6 \times 0.25$

*Three term equation of motion for particle,
with correct use of g*

M1

Correct equation

A1ft

$$T = 57.3$$

ag *Correct T from correct working*

A1ft

3

(c) $57.3 - F = 10 \times 0.25$

Three term equation of motion for the block

M1

Correct equation

A1ft

$$F = 54.8$$

Correct F

A1ft

$$R = 10 \times 9.8 = 98$$

R = 98 seen in working

B1

$$54.8 = 98\mu$$

Use of $F = \mu R$

m1

$$\mu = \frac{54.8}{98} = 0.559$$

correct μ

A1ft

6

[11]

Q4.

(a) $R = 60 \times 9.8 \times \cos 40^\circ = 450 \text{ N}$

Resolving perpendicular to the plane

M1

Correct reaction force

A1

2

(b) $F = 60 \times 9.8 \times \cos 50^\circ = 378$

Resolving parallel to the plane

M1

Correct F

A1

$$F \leq \mu R$$

Use of $F \leq \mu R$ or $F = \mu R$

m1

$$\mu \geq 0.839$$

Correct inequality

Accept 0.840

A1

4

(c) $F = 0.2 \times 60 \times 9.8 \cos 40^\circ$

$$60a = 60 \times 9.8 \cos 50^\circ - 0.2 \times 60 \times 9.8 \cos 40^\circ$$

Use of $F = \mu R$ to find F but not with $R = 60g$

M1

Three term equation of motion

M1

Correct equation

A1

$$a = 4.80 \text{ ms}^{-2}$$

Correct acceleration

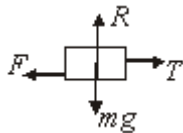
A1

5

[11]

Q5.

(a)



Correct diagram

B1

1

(b) $R = 4 \times 9.8 = 39.2 \text{ N}$

Correct normal reaction

B1

1

(c) $T - 4 \times 39.2 = 4 \times 2$

Three term equation of motion

M1

Correct equation

A1

$$T = 23.7 \text{ N (to 3sf)}$$

Correct T

A1

3

(d) $20 - 0.4 \times 39.2 = 4a$

M1

$a = 1.08 \text{ ms}^{-2}$

A1

2

[7]

Q6.

(a) $12g - T = 12a$

Equation of motion for one particle

M1

Correct equation

A1

$T - 8g = 8a$

Equation of motion for other particle

M1

Correct equation

A1

$8a + 8g = 12g - 12a$

$20a = 4g$

$a = \frac{4g}{20} = 1.96 \text{ ms}^{-2}$

ag *Correct a from correct working*

A1

5

(b) $T = 8 \times 1.96 + 8 \times 9.8$

$= 94.1 \text{ N}$

Substituting value for a into equation of motion to find T

M1

Correct T

A1

2

(c) $7 = 0 + 1.96t$

Using constant acceleration to find t

M1

$$t = \frac{7}{1.96} = 3.57 \text{ s (to 3 sf)}$$

Correct t

A1

2

[9]

Q7.

(a) $T_1 = 4 \times 9.8 = 39.2 \text{ N}$

Correct tension in left string

B1

$$T_2 = 3 \times 9.8 = 29.4 \text{ N}$$

Correct tension in right string

B1

2

(b) $39.2 \sin 43^\circ = 29.4 \sin \theta$

Resolving horizontally

M1

$$\sin \theta = \frac{39.2 \sin 43^\circ}{29.4} = 0.9093$$

Correct equation

A1

Correct expression for $\sin \theta$

A1

$$\theta = 65.4^\circ$$

Finding θ

M1

ag *Correct θ from correct working*

A1

5

(c) $9.8m = 39.2 \cos 43^\circ + 29.4 \cos \theta$

Resolving vertically

M1

Correct equation

A1

$$m = \frac{39.2 \cos 43^\circ + 29.4 \cos \theta}{9.8} = 4.17 \text{ kg}$$

Finding m

M1

Correct m

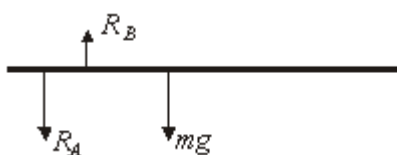
A1

4

[11]

Q8.

(a)



Correct force diagram

B1

1

(b) $0.8R_A = 0.5 \times 40 \times 9.8$

Moment equation to find R_A

M1

Correct equation

A1

$$R_A = \frac{0.5 \times 40 \times 9.8}{0.8} = 245 \text{ N}$$

ag *Correct reaction from correct working*

A1

3

(c) $R_B = 245 + 40 \times 9.8 = 637 \text{ N}$

Use of equilibrium to form an equation

M1

Correct reaction

A1

2

(d) $0.8R_A = 0.5 \times 40 \times 9.8 + 3 \times 5 \times 9.8$

Four term moment equation

M1

Correct equation

A1

$$R_A = \frac{0.5 \times 40 \times 9.8 + 3 \times 5 \times 9.8}{0.8}$$

Finding reaction

M1

$$= 429 \text{ N (to 3sf)}$$

Correct reaction

A1

$$R_B = 429 + 40 \times 9.8 + 5 \times 9.8$$

Equation to find other reaction

M1

$$= 870 \text{ N (to sf)}$$

Correct reaction

A1

6

[12]

Q9.

(a) $\mathbf{F} = (3\mathbf{i} + 4\mathbf{j}) + (6\mathbf{i} - 8\mathbf{j})$

Addition of the two forces

M1

2

$$= 9\mathbf{i} - 4\mathbf{j}$$

Correct resultant

A1

(b) $F = \sqrt{9^2 + 4^2} = 9.85 \text{ N}$

Finding magnitude

M1

Correct magnitude

A1

2

(c) $\tan \alpha = \frac{4}{9}$

Using tan to find the angle

M1

Correct equation

A1

$\alpha = 24.0^\circ$

Correct angle

A1

3

[7]

Q10.

(a) $0.8 = \frac{1}{2} a \times 2^2$

Constant acceleration equation with $u = 0$

M1

Correct equation

A1

$a = 0.4 \text{ ms}^{-1}$

Correct acceleration

A1

3

(b) $2 \times 0.4 = 2 \times 9.8 - T$

Three term equation of motion

M1

Correct equation

A1

$T = 2 \times 9.8 - 2 \times 0.4 = 18.8 \text{ N}$

Correct tension

A1

3

(c) $0.4m = 18.8 - 9.8m$

Three term equation of motion

M1

Correct equation
Solving for m

A1

$$m = \frac{18.8}{10.2} = 1.8 \text{ kg}$$

M1

Correct m (2sf)

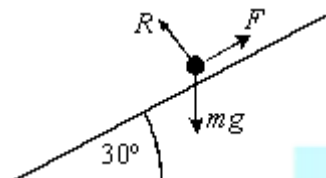
A1

4

[10]

Q11.

(a)



Correct force diagram

B1

1

(b) $R = 5 \times 9.8 \cos 30^\circ$

Finding normal reaction

M1

$$F = 0.2 + 5 \times 9.8 \cos 30^\circ = 8.5 \text{ N (to sf)}$$

Finding friction

M1

Correct friction

A1

3

(c) $5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ = 5a$

Three term equation of motion

M1

Correct equation

A1

$$a = \frac{5 \times 9.8 \sin 30^\circ - 0.2 \times 5 \times 9.8 \cos 30^\circ}{5}$$

Solving for a

M1

$$= 3.20 \text{ ms}^{-2}$$

Correct a

A1

4

(d) $v^2 = 0^2 + 2 \times 3.203 \times 1.2$

Constant acceleration equation with $u = 0$

M1

Correct equation

A1 ft.

$$v = 2.77 \text{ ms}^{-1}$$

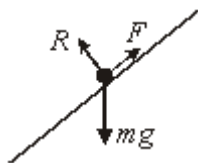
A1 ft.

3

[11]

Q12.

(a)



Correct force diagram

No extra forces

B1

1

(b) (i) $R = 80 \times 9.8 \cos 30^\circ = 679.0 \text{ N}$

Resolving perpendicular to the slope

Correct Reaction force

M1 A1

2

(ii) $F = 0.3R = 204 \text{ N}$

Multiply by 0.3

ag Correct answer obtained

M1 A1

2

(c) $80a = 80 \times 9.8 \sin 30^\circ - 203.7$

Three term equation of motion
Correct equation

M1 A1

$$a = 2.35 \text{ ms}^{-2}$$

Solving for a
awrt 2.35

m1 A1

4

[9]

Q13.

$$4R_B = 2.2 \text{ m}$$

Moments about A or B
Correct equation
Correct Reaction force

M1 A1

$$R_B = 216 \text{ N (to 3sf)}$$

A1

$$R_A + R_B = 40 \times 9.8$$

Moments about other end or resolve vertically

M1

$$R_A = 176 \text{ N (to 3 sf)}$$

Other reaction correct

A1

[5]

Q14.

(a) $T \cos 42^\circ = 60 \cos 40^\circ$

Resolving horizontally

M1

Correct equation

A1

$$T = \frac{60 \cos 40^\circ}{\cos 42^\circ} = 61.8 \text{ N (to 3sf)}$$

awrt 61.8

A1

3

(b) $T \sin 42^\circ + 60 \sin 40^\circ = 9.8m$

Resolving vertically with three terms

M1

Correct equatino with T from bove

A1

$$m = \frac{T \sin 42^\circ + 60 \sin 40^\circ}{9.8} = 8.16 \text{ kg (to 3sf)}$$

Solving for m

m1

awrt 8.16

A1

4

[7]

Q15.

(a) (i) $60 \times 9.8 - T = 60a$

Equation of motion for one boy

Correct equation

M1 A1

$$T - 40 \times 9.8 = 40a$$

Equation of motion for other boy

Correct equation

M1

$$588 - (40a + 392) = 60a$$

$$a = 1.96 \text{ ms}^{-2}$$

ag Correct answer from correct working

A1

5

(ii) $T = 40 \times 1.96 + 392 = 470 \text{ N (to 3 sf)}$

Substituting for a and solving for T

awrt 470

M1 A1 M1

2

(b) $v^2 = 0^2 + 2 \times 1.96 \times 2 = 7.84$

Finding v^2 using $s = 2$, $u = 0$ and $a = 1.96$

Correct v or v^2

M1 A1

$0^2 = 7.84 + 2 \times (-9.8)s$

Equation for s with $v = 0$ and $a = \pm g$

m1 A1

$s = 0.4$

Solving for s

m1

$h_{\max} = 2 + 0.4 = 2.4 \text{ m}$

Solving for s

Obtaining 0.4

Adding 2 to get 2.4

m1 A1 A1

7

[14]

Q16.

(a) $\mathbf{F} = (2 + 6 - 2)\mathbf{i} + (3 - 8 + 10)\mathbf{j}$

Summing the forces

M1

$= 6\mathbf{i} + 5\mathbf{j}$

Correct resultant

A1

2

(b) $\mathbf{a} = \frac{1}{8} (6\mathbf{i} + 5\mathbf{j}) = 0.75\mathbf{i} + 0.625\mathbf{j}$

Dividing by 8 to get correct value

B1

1

(c) (i) $\mathbf{v} = (3\mathbf{i} - 2\mathbf{j}) + 40(0.75\mathbf{i} + 0.625\mathbf{j})$

use of $\mathbf{v} = \mathbf{u} + \mathbf{at}$

M1

$= 33\mathbf{i} + 23\mathbf{j}$

Correct $\mathbf{v}$

A1

2

(ii) $\mathbf{r} = (3\mathbf{i} - 2\mathbf{j}) \times 40 + \frac{1}{2} (0.75\mathbf{i} + 0.625\mathbf{j}) \times 40^2$
Vector constant acceleration equation to find r
Correct equation

M1 A1

$= 720\mathbf{i} + 420\mathbf{j}$
Correct position vector

A1

$r = \sqrt{720^2 + 420^2} = 834 \text{ m}$
Finding magnitude
Correct magnitude

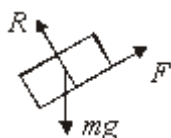
M1 A1

5

[10]

Q17.

(a)



Correct force diagram

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B1

1

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(b) $R = 7 \times 9.8 \cos 30^\circ = 59.4 \text{ N}$
Resolving perpendicular to plane
Correct force

M1/A1

2

(c) $F = 7 \times 9.8 \sin 30^\circ = 34.3 \text{ N}$
Resolving parallel to plane
Correct force

M1/A1

2

(d) $34.3 \leq \mu \times 59.4$
Using $F = \mu R$ or $F \leq \mu R$

M1

$$\mu \geq 0.577$$

Correct inequality

A1

2

- (e) Remains at rest, as mass appears in both F and R , so inequality unchanged.

Conclusion

B1

Reason

B1

2

[9]

Q18.

(a) $T - 0.8 \times 5 \times 9.8 = 5a$

Equation of motion for one particle

M1

$$T = 5a + 39.2$$

Correct equation

A1

$$3 \times 9.8 - T = 3a$$

Equation of motion for other particle

M1

$$29.4 - (5a + 39.2) = 3a$$

Correct equation

A1

$$-9.8 = 8a$$

Solving for a

M1

$$a = -1.225 \text{ ms}^{-2}$$

Correct a

A1

6

(b) (ii) $T = 5a + 39.2$

Substituting and solving for T

M1

$$T = 33.1\text{N}$$

Correct T

A1

2

(c) $0^2 = 2^2 + 2 \times (-1.225)s$

Use of CA equation with $v = 0$

Correct equation (allow $a = 1.225$)

M1/A1

$$s = \frac{4}{2.45} = 1.63 \text{ m}$$

*Correct answer from correct working
(allow 1.64)*

A1

3

[11]



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